Reason about transformations

Use the scaffold, show each step and ask for an adult check when marked.

Name: ______________________________ Date: ______________

1. Read the model: A point becomes (7, -4) after moving 9 right and 3 down. Find its start. Copy or complete the first step.

2. Plot A(-4, 3). Which quadrant contains A?

3. Point P moves from (2, -5) to (-3, 1). Describe the translation.

4. Miro says: Miro changes x and y in the same way without checking the direction or mirror line. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: A point becomes (7, -4) after moving 9 right and 3 down. Find its start. Copy or complete the first step.
   Answer (-2, -1).

2. Plot A(-4, 3). Which quadrant contains A?
   Answer Quadrant II.

3. Point P moves from (2, -5) to (-3, 1). Describe the translation.
   Answer 5 left and 6 up.

4. Miro says: Miro changes x and y in the same way without checking the direction or mirror line. Is this correct? Explain.
   Answer Track horizontal and vertical changes separately and apply the exact transformation to every point.

Reason about transformations

Work independently. Show a complete method and check at least one answer.

Name: ____ Date: __________________

1. A point becomes (7, -4) after moving 9 right and 3 down. Find its start.

2. Plot A(-4, 3). Which quadrant contains A?

3. Point P moves from (2, -5) to (-3, 1). Describe the translation.

4. Solve this independently and show every stage: A point becomes (7, -4) after moving 9 right and 3 down. Find its start.

5. Use a second method or representation for: Plot A(-4, 3). Which quadrant contains A?

6. Check this result using an inverse, estimate or known fact: Point P moves from (2, -5) to (-3, 1). Describe the translation.

7. Prove or explain your reasoning: Translate (4, -2) by 3 left and 5 up.

8. Identify and correct this misconception: Miro changes x and y in the same way without checking the direction or mirror line.

9. Write a complete sentence explaining how you checked one answer.

10. Write and solve a similar question to: Point P moves from (2, -5) to (-3, 1). Describe the translation.

Teacher answers and checking prompts

1. A point becomes (7, -4) after moving 9 right and 3 down. Find its start.
   Answer (-2, -1).

2. Plot A(-4, 3). Which quadrant contains A?
   Answer Quadrant II.

3. Point P moves from (2, -5) to (-3, 1). Describe the translation.
   Answer 5 left and 6 up.

4. Solve this independently and show every stage: A point becomes (7, -4) after moving 9 right and 3 down. Find its start.
   Answer (-2, -1).

5. Use a second method or representation for: Plot A(-4, 3). Which quadrant contains A?
   Answer Expected result: Quadrant II. Method or representation may vary.

6. Check this result using an inverse, estimate or known fact: Point P moves from (2, -5) to (-3, 1). Describe the translation.
   Answer Expected result: 5 left and 6 up. A valid check must be shown.

7. Prove or explain your reasoning: Translate (4, -2) by 3 left and 5 up.
   Answer (1, 3). The explanation must show why the method works.

8. Identify and correct this misconception: Miro changes x and y in the same way without checking the direction or mirror line.
   Answer Track horizontal and vertical changes separately and apply the exact transformation to every point.

9. Write a complete sentence explaining how you checked one answer.
   Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.

10. Write and solve a similar question to: Point P moves from (2, -5) to (-3, 1). Describe the translation.
    Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Reason about transformations

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: ____ Date: __________________

1. Prove or explain your reasoning: Translate (4, -2) by 3 left and 5 up.

2. Prove the answer to this question, rather than only calculating it: A point becomes (7, -4) after moving 9 right and 3 down. Find its start.

3. Solve this in two different ways and compare them: Plot A(-4, 3). Which quadrant contains A?

4. Work backwards from the answer to reconstruct the question: 5 left and 6 up.

5. Explain why this reasoning fails: Miro changes x and y in the same way without checking the direction or mirror line.

6. Create a new example that tests the same idea as: Point P moves from (2, -5) to (-3, 1). Describe the translation.

7. Find a second method or representation. Compare its efficiency with your first method.

8. Write one always, sometimes or never statement about today's learning and justify it.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Prove or explain your reasoning: Translate (4, -2) by 3 left and 5 up.
   Answer (1, 3). The explanation must show why the method works.

2. Prove the answer to this question, rather than only calculating it: A point becomes (7, -4) after moving 9 right and 3 down. Find its start.
   Answer Expected result: (-2, -1). Proof or justification must match the lesson structure.

3. Solve this in two different ways and compare them: Plot A(-4, 3). Which quadrant contains A?
   Answer Expected result: Quadrant II. Two valid methods and a comparison are required.

4. Work backwards from the answer to reconstruct the question: 5 left and 6 up.
   Answer One valid reconstruction is: Point P moves from (2, -5) to (-3, 1). Describe the translation. Other equivalent questions are acceptable.

5. Explain why this reasoning fails: Miro changes x and y in the same way without checking the direction or mirror line.
   Answer Track horizontal and vertical changes separately and apply the exact transformation to every point.

6. Create a new example that tests the same idea as: Point P moves from (2, -5) to (-3, 1). Describe the translation.
   Answer Answers vary. The example and solution must preserve the mathematical structure.

7. Find a second method or representation. Compare its efficiency with your first method.
   Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.

8. Write one always, sometimes or never statement about today's learning and justify it.
   Answer Answers vary. A valid example and counterexample should support the classification.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
   Answer Answers vary. The error must be mathematically relevant and the correction must be precise.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
    Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.